Worksheet 15: Optimization I

MATH 1271

10/29/2026

1. Suppose that $g(x)$ is a function that is continuous at every value of x except at $x = 2$. You may assume that this is because g has a vertical asymptote at $x = 2$.

   You may also assume that the **first derivative** of g is

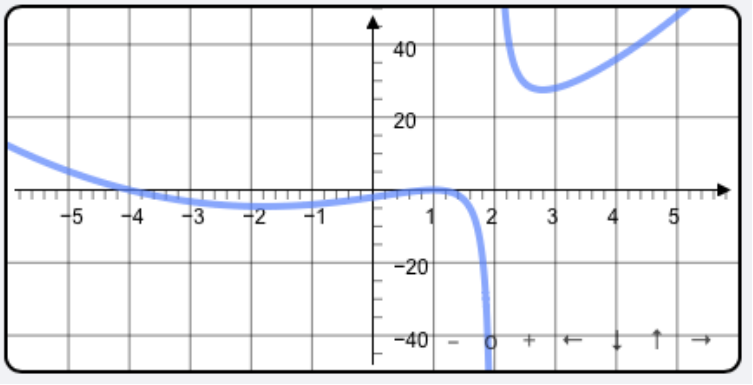
   $$g'(x) = \frac{(x+4)(x-1)^2}{(x-2)}$$

   (a) Determine all critical numbers of g.

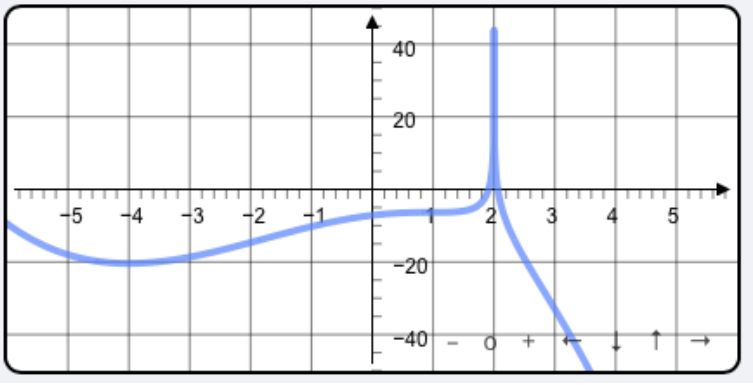
   (b) By developing a carefully labeled first derivative sign chart, decide whether g has a local minimum, local maximum, or neither at each critical number.

(c) Which of the following graphs is a possible graph of $y = g(x)$?

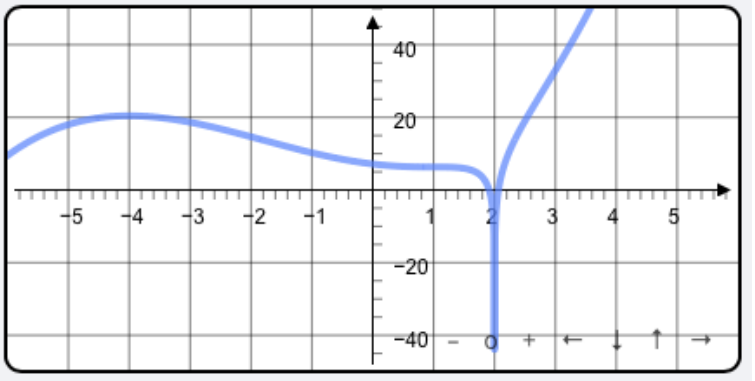
i.



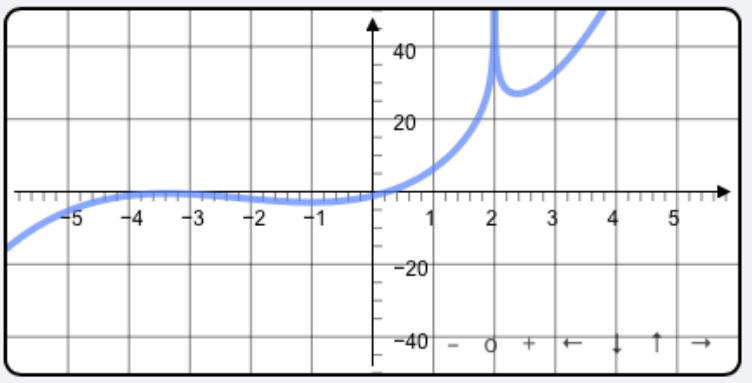
ii.



iii.



iv.



2. (a) Let $f(x) = x^4(x-1)^3$.

i. The function f has three critical numbers; find them.

ii. You may assume that

$$f''(x) = 6x^2(x-1)(7x^2 - 8x + 2)$$

What does the Second Derivative Test tell you about the critical numbers you found in part (a)?

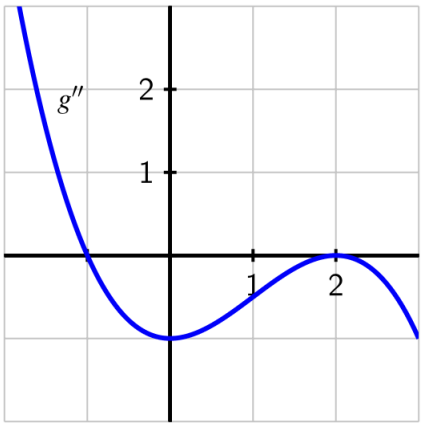
iii. What does the First Derivative Test tell you about the critical numbers you found in part (a)?

(b) Let $g(t) = t^2 \ln(t)$.

i. The function g has one critical number; what is it?

ii. Is this critical number a local maximum of g, a local minimum of g, or neither one?

3. Suppose that g is a function whose **second derivative**, g'', is given by the graph below:



(a) Find the x-coordinates of all points of inflection of g.

(b) Suppose you are given that $g'(-1.67857351) = 0$. Is there a local maximum, local minimum, or neither (for the function g) at this critical point, or is it impossible to say?

(c) Assuming that $g''(x)$ is a polynomial (and that all important behavior of $g''(x)$ is seen in the graph above), what degree polynomial do you think $g(x)$ is? (Hint: Think about the Power Rule. How does the degree of a polynomial relate to the degree of its second derivative?)